

# Mathletes Formulas

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# 1. Polynomials

## 1.1. Linear Equations

For polynomials of degree one, the variable  $m$  is used to refer to the slope of the lines drawn by their graphs. Parallel lines have equivalent slopes, while perpendicular lines have slopes that are reciprocals of one another ( $m \perp \frac{1}{m}$ ).

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

**Slope-Intercept Form**  $y = mx + b$

**Point-Slope Form**  $y - y_1 = m(x - x_1)$

**Point-Point Form**

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

## 1.2. Quadratics

**Quadratic Equation**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If you let  $D = b^2 - 4ac$

1.  $D > 0 \implies 2$  real solutions
2.  $D = 0 \implies 1$  real solution
3.  $D < 0 \implies$  no real solutions

**Minimum and Maximum Values**

Consider the function  $f(x) = ax^2 + bx + c$  with vertex  $(-\frac{b}{2}a, f(-\frac{b}{2}a))$

1.  $a > 0 \implies f$  has a *minimum* at  $x = -\frac{b}{2}a$  The minimum value is  $f(-\frac{b}{2}a)$
2.  $a < 0 \implies f$  has a *maximum* at  $x = -\frac{b}{2}a$  The maximum value is  $f(-\frac{b}{2}a)$

## 1.3. General Polynomials

Let  $n$  be a non-negative integer and let  $a_n, a_{n-1}, \dots, a_2, a_1, a_0$  be real numbers such that  $a_n \neq 0$ . The function:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

is a polynomial function of  $x$  with degree  $n$ .

**Linear Factorization Theorem**

If  $f(x)$  is a polynomial of degree  $n$ , where  $n > 0$ , then  $f(x)$  has precisely  $n$  linear factors

$$f(x) = a_n(x - c_1)(x - c_2)\dots(x - c_n)$$

where  $c_1, c_2, \dots, c_n$  are complex numbers.

### Rational Zero Test

If the polynomial has *integer* coefficients, then every rational zero of  $f$  has the form  $\frac{p}{q}$ , where  $p$  and  $q$  have no common factors other than 1, and

$$p = \text{a factor of the constant term } a_0 \qquad q = \text{a factor of the leading coefficient } a_n$$

### Conjugate Pairs

Let  $f$  be a polynomial function with *real coefficients*. If  $a + bi$ , where  $b \neq 0$ , is zero of  $f$ , the complex conjugate  $a - bi$  is also a zero of  $f$ .

For polynomials with *integer coefficients*,  $a + \sqrt{b}$  being a zero implies  $a - \sqrt{b}$  is also a zero.

### Roots

Let  $f$  be the polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

$$\text{The sum of all roots} = -\frac{a_n}{a_{n-1}}$$

$$\text{The product of all roots} = (-1)^n \frac{a_0}{a_n}$$

## 1.4. Rational Functions

Let  $f$  be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_n x^n + b_{n-1} x^{n-1} + \dots + b_1 x + b_0}$$

where  $N(x)$  and  $D(x)$  have no common factors.

1. The graph of  $f$  has *vertical* asymptotes at the zeros of  $D(x)$ .
2. The graph of  $f$  has at most one *horizontal asymptote* determined by comparing the degrees of  $N(x)$  and  $D(x)$ .
  - a.  $n < 0 \implies$  the line  $y = 0$  is the horizontal asymptote.
  - b.  $n = 0 \implies$  the line  $y = \frac{a_n}{b_n}$  is the horizontal asymptote.
  - c.  $n > 0 \implies$  there is no horizontal asymptote.

Suppose  $N(x)$  and  $D(x)$  had some common factor  $(x - c)$ . The graph of  $f$  would then have a hole at  $(c, f(c))$

## 2. Trigonometry

### 2.1. Functions

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos x = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan x = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin x}{\cos x}$$

$$\csc x = \frac{1}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\cot x = \frac{1}{\tan x}$$

| Function    | Domain       | Range                   |
|-------------|--------------|-------------------------|
| $\sin^{-1}$ | $[-1, 1]$    | $[-90^\circ, 90^\circ]$ |
| $\cos^{-1}$ | $[-1, 1]$    | $[0^\circ, 180^\circ]$  |
| $\tan^{-1}$ | $\mathbb{R}$ | $[-90^\circ, 90^\circ]$ |
| $\cot^{-1}$ | $\mathbb{R}$ | $[0^\circ, 180^\circ]$  |

|     | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ | $180^\circ$ | $270^\circ$ | $360^\circ$ |
|-----|-----------|----------------------|----------------------|----------------------|------------|-------------|-------------|-------------|
| sin | 0         | 1                    | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1          | 0           | -1          | 0           |
| cos | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$        | 0          | -1          | 0           | 1           |
| cot | 0         | $\frac{\sqrt{3}}{3}$ | 1                    | $\sqrt{3}$           | undef      | 0           | undef       | 0           |

### 2.2. Identities

#### Radians and Degrees

$$n^\circ = \frac{\pi}{180} n \text{ rad}$$

$$m \text{ rad} = \left( \frac{180}{\pi} m \right)^\circ$$

#### Co-function

$$\cos x = \sin(90 - x)$$

$$\sin x = \cos(90 - x)$$

$$\tan x = \cot(90 - x)$$

#### Pythagorean

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

#### Negative

$$\sin(-x) = -\sin x \quad \csc(-x) = -\csc x$$

$$\cos(-x) = \cos x \quad \sec(-x) = \sec x$$

$$\tan(-x) = -\tan x \quad \cot(-x) = -\cot x$$

#### Sum to Difference

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \pm \sin x \sin y$$

$$\tan(x \pm y) = \frac{\sin(x \pm y)}{\cos(x \pm y)}$$

#### Half-Angle

$$\sin\left(\frac{x}{2}\right) = \pm \sqrt{\frac{1 - \cos x}{2}} *$$

$$\cos\left(\frac{x}{2}\right) = \pm \sqrt{\frac{1 + \cos x}{2}} *$$

$$\tan\left(\frac{x}{2}\right) = \pm \frac{\sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right)} *$$

#### Double-Angle

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{\sin 2x}{\cos 2x}$$

\* where you consider the angle's location to determine the sign

### Sum to Product

$$\begin{aligned}\sin x \pm \sin y &= 2 \sin\left(\frac{x \pm y}{2}\right) \cos\left(\frac{x \mp y}{2}\right) & \tan x \pm \tan y &= \frac{\sin(x \pm y)}{\cos x \cos y}, x \neq 90^\circ \text{ and } 270^\circ \\ \cos x + \cos y &= 2 \cos\left(\frac{x + y}{2}\right) \cos\left(\frac{x - y}{2}\right) & \cot x \pm \cot y &= \frac{\sin(y \pm x)}{\sin x \sin y}, x \neq 180^\circ \text{ and } x \neq 360^\circ \\ \cos x - \cos y &= -2 \sin\left(\frac{x + y}{2}\right) \sin\left(\frac{x - y}{2}\right)\end{aligned}$$

(1) where  $x \neq 90^\circ$  and  $x \neq 270^\circ$

(2) where  $x \neq 180^\circ$  and  $x \neq 360^\circ$

### Product to Sum

$$\begin{aligned}\sin x \sin y &= \frac{\cos(x - y) - \cos(x + y)}{2} \\ \cos x \cos y &= \frac{\cos(x + y) + \cos(x - y)}{2} \\ \sin x \cos y &= \frac{\sin(x + y) + \sin(x - y)}{2}\end{aligned}$$

## 2.3. Laws

### Law of Sines

$$\begin{aligned}2r &= \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \\ \text{area} &= \frac{1}{2}ab \sin A\end{aligned}$$

### Law of Cosines

$$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos A \\ \cos A &= \frac{b^2 + c^2 - a^2}{2bc}\end{aligned}$$

## 3. Graphs and Geometry

### 3.1. Lines and Points

#### Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

#### Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

#### Perpendicular Bisector

$$2(x_2 - x_1)x + 2(y_2 - y_1)y = x_2^2 + y_2^2 - x_1^2 - y_1^2$$

#### Pythagorean Theorem

$$a^2 + b^2 = c^2 \implies c = \sqrt{a^2 + b^2}$$

### 3.2. Shapes

#### Circles

$$(x - m)^2 + (y - n)^2 = r^2$$

$$A = \pi r^2$$

$$C = 2\pi r$$

## Regular Polygons

$$\text{Sum of Interior Angles} = 180(n-2)^\circ \quad A = \frac{1}{2} \cdot a \cdot P, \text{ where } P = n \cdot s \text{ and } a = \text{distance from the center to the middle of a side}$$

## Volumes

$$V_{\text{Cylinder}} = \pi r^2 h \quad V_{\text{Cone}} = \frac{1}{3} \pi r^2 h \quad V_{\text{Sphere}} = \frac{4}{3} \pi r^3$$

## Triangle Similarity

Two triangles  $\triangle ABC$  and  $\triangle DEF$  are similar if

|                                                 |                                                           |                                                 |
|-------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------|
| <b>SSS</b>                                      | <b>SAS</b>                                                | <b>AA</b>                                       |
| $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$ | $\frac{AB}{DE} = \frac{AC}{DF}$ and $\angle A = \angle D$ | $\angle A = \angle D$ and $\angle B = \angle E$ |

## 3.3. Graph Manipulation

| Transformation      | Conditions | Description                            |
|---------------------|------------|----------------------------------------|
| $g(x) = f(x \pm c)$ | (+)        | shift left $c$ units                   |
|                     | (-)        | shift right $c$ units                  |
| $g(x) = f(x) \pm c$ | (+)        | shift left $c$ units                   |
|                     | (-)        | shift right $c$ units                  |
| $g(x) = cf(x)$      | $c < 0$    | vertical compress by a factor of $c$   |
|                     | $c > 0$    | vertical stretch by a factor of $c$    |
| $g(x) = f(cx)$      | $c < 0$    | horizontal stretch by a factor of $c$  |
|                     | $c > 0$    | horizontal compress by a factor of $c$ |
| $g(x) = -f(x)$      |            | reflect about the x-axis               |
| $g(x) = f(-x)$      |            | reflect about the y-axis               |

## 4. Miscellaneous

### 4.1. Inverse Functions

Let  $f$  and  $g$  be two function such that

$$f(g(x)) = x \text{ for all } x \text{ in the domain of } g \quad g(f(x)) = x \text{ for all } x \text{ in the domain of } f$$

Under these conditions, the function  $g$  is the *inverse function* of the function  $f$  (and vice-versa). The function  $g$  can then be denoted as  $f^{-1}$ . So,

$$f(f^{-1}(x)) = x \quad f^{-1}(f(x)) = x$$

The domain of  $f$  must be equal to the range of  $f^{-1}$ , and the range of  $f$  must be equal to the domain of  $f^{-1}$ .

## 4.2. Complex Numbers

Let  $a$  and  $b$  be real numbers. The number  $a + bi$  is a *complex number* written in *standard form*. The number  $a$  is the *real part* and the number  $bi$  is the *imaginary part* of the complex number.

When  $b = 0$ , the number  $a + bi$  is a real number. When  $b \neq 0$ , the number  $a + bi$  is an *imaginary number*. A number of the form  $bi$ , where  $b \neq 0$ , is a *pure imaginary number*.

$$i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, i^n = i^{n \bmod 4}$$

## 4.3. Relationships

### Exponential

$$\begin{array}{ll} a^0 = 1 & \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \\ a^m \cdot a^n = a^{m+n} & (a^m)^n = a^{m \cdot n} \\ \frac{a^m}{a^n} = a^{m-n} & a^{\frac{1}{n}} = \sqrt[n]{a} \\ (ab)^n = a^n \cdot b^n & a^m = a^n \implies m = n \end{array}$$

### Logarithmic

$$\begin{array}{ll} \log_a 1 = 0 & \log ab^n = n \log ab \\ \log_a a = 1 & \log a^n b = \frac{1}{n} \log ab \\ \log x = \log_{10} x & \log_n ab = \log_n a + \log_n b \\ \ln x = \log_e x & \log_n \frac{a}{b} = \log_n a - \log_n b \\ y = \log_b x \implies b^y = x & \log_n a = \log_n b \implies a = b \\ a^{\log_a b} = b & \end{array}$$